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Abstract—The rapid progress of deep learning has made it possible to generate synthetic voices that closely resemble real human speech, leading to the rise of deepfake audio. These artificially produced voices pose significant risks, including impersonation, fraud, misinformation, and identity misuse. This paper presents a deep learning-based system for detecting deepfake audio using Mel-Frequency Cepstral Coefficients (MFCC) and a Long Short-Term Memory (LSTM) neural network. The proposed model analyzes temporal variations and frequency characteristics of speech to distinguish between real and AI-generated audio. A web-based interface allows users to upload audio files for real-time predictions. Experimental results demonstrate high accuracy, strong robustness, and reliable performance across diverse audio samples. The system provides a practical and scalable solution for audio authentication in cybersecurity, media verification, and digital forensics.

1. Introduction

Advances in deep learning have enabled artificial voice generation models to replicate human speech with exceptional accuracy. These systems capture vocal tone, pitch, rhythm, and accent, producing audio that appears genuine to human listeners. While such technologies support virtual assistants, text-to-speech systems, and voice-based accessibility tools, they also create new challenges in digital security.

Deepfake audio can be misused to manipulate communication, perform impersonation attacks, deceive financial institutions, and spread misinformation. Multiple incidents have been reported where criminals used synthetic voice cloning to execute fraudulent transactions. Traditional audio verification systems fail to detect synthetic manipulation because deepfake generation removes many artifacts that older systems depend on. Therefore, advanced approaches based on temporal and frequency analysis are needed. LSTM models are well-suited for this purpose as they learn sequential patterns in audio that reveal subtle inconsistencies.

This project focuses on building a lightweight, real-time deepfake detection system using MFCC and LSTM. The aim is to deliver a practical, accurate, and user-friendly system suitable for real-world audio authentication.

Deepfake audio has emerged as one of the most dangerous forms of digital manipulation because it directly targets human trust. Unlike visual deepfakes, which often show visible flaws, audio deepfakes can deceive individuals even in low-quality communication environments such as phone calls. This makes them ideal for social engineering, political disruption, and identity-based attacks. Voice-based authentication systems used in banking, smart device control, and customer support are highly vulnerable to spoofing through synthetic audio. As companies continue to rely on voice biometrics, detecting manipulated audio becomes essential to prevent unauthorized access and data breaches.

Recent advancements in generative models such as Tacotron, WaveNet, VITS, and diffusion-based audio synthesizers have dramatically increased the realism of synthetic speech. These models can clone a person's voice using only a few seconds of audio, making deepfake creation more accessible than ever.

2. Literature Review

Deepfake audio detection has become increasingly important as neural speech synthesis tools improve in realism and accessibility. Recent studies mainly focus on enhancing feature extraction, improving robustness against adversarial attacks, and strengthening temporal modeling for detecting artificially generated speech. This section reviews key research contributions relevant to the development of our MFCC-LSTM deepfake audio detection system.

Traditional anti-spoofing systems rely heavily on spectral and cepstral representations of speech. Todisco et al. introduced the Constant-Q Cepstral Coefficients (CQCC), demonstrating their effectiveness in identifying spectral artifacts produced by synthetic and converted speech signals [1]. Their work established the importance of high-resolution time-frequency analysis in spoof detection and motivated the use of lightweight cepstral features such as MFCC for real-time systems.

Recent research has explored fusion-based acoustic representations. Kumar and Kavitha proposed a multi-feature fusion approach combining MFCC, spectral contrast, zero-crossing rate, and pitch-based descriptors to improve classification performance [2]. Although feature fusion enhances

accuracy, it significantly increases computational cost. Considering real-time deployment constraints, our system adopts MFCC alone due to its balance of discriminative power and efficiency.

Deepfake audio detectors have also been shown to be vulnerable to adversarial perturbations. Liu et al. examined the susceptibility of audio deepfake detection models to adversarial attacks and demonstrated that even small perturbations can mislead state-of-the-art systems [3]. While adversarial training enhances robustness, it requires extensive data and computational resources. Therefore, our design avoids adversarial augmentation and prioritizes robust preprocessing techniques to increase real-world performance.

Hybrid deep-learning models combining convolutional and recurrent architectures have gained attention due to their ability to capture both spectral and temporal cues. Asuai et al. introduced a CNN-LSTM-based hybrid model that integrates spectrogram feature extraction with temporal sequence learning, achieving competitive performance across benchmark datasets [4]. However, such systems require high computational resources, making them less suitable for lightweight deployments. Inspired by their findings, we utilize an LSTM-only architecture, which maintains strong temporal modeling capabilities with reduced complexity.

Several works emphasize spectrogram-based deepfake detection. Pham et al. proposed an ensemble learning framework using multiple spectrogram representations, highlighting the role of diverse frequency encodings in detecting synthetic artifacts [5]. Spectrogram-based CNNs achieve strong results in controlled environments but often suffer from overfitting on smaller datasets and struggle with long-term temporal dependencies. To overcome these issues, our system uses a large, balanced dataset and MFCC features that preserve essential speech characteristics while remaining compact.

Robust preprocessing has also been recognized as a key factor for generalization in real-world scenarios. Chakravarty and Dua demonstrated that noise reduction, resampling, and normalization significantly improve model performance on noisy and variable audio samples [6]. Motivated by their findings, our system incorporates noise reduction, silence trimming, mono conversion, and 16 kHz resampling to enhance consistency across audio inputs.

Overall, existing research highlights the need for lightweight, temporally aware, and preprocessing-robust systems for real-time deepfake audio detection. These insights directly shaped the design of our MFCC-based LSTM model.

3. Methodology

The proposed deepfake audio detection system follows a structured and modular pipeline that transforms raw speech input into a classification output indicating whether the audio is REAL or FAKE. The methodology consists of five major stages: Preprocessing, Feature Extraction, Model Architecture, System Backend, and Frontend Integration. Each stage plays a crucial role in ensuring accurate and real-time detection of synthetic speech.

Preprocessing prepares the raw audio for consistent analysis by eliminating variations caused by different recording environments or devices. The system performs the following operations: All audio files are resampled to 16 kHz, which is a standard sampling rate for speech processing. This ensures uniform temporal

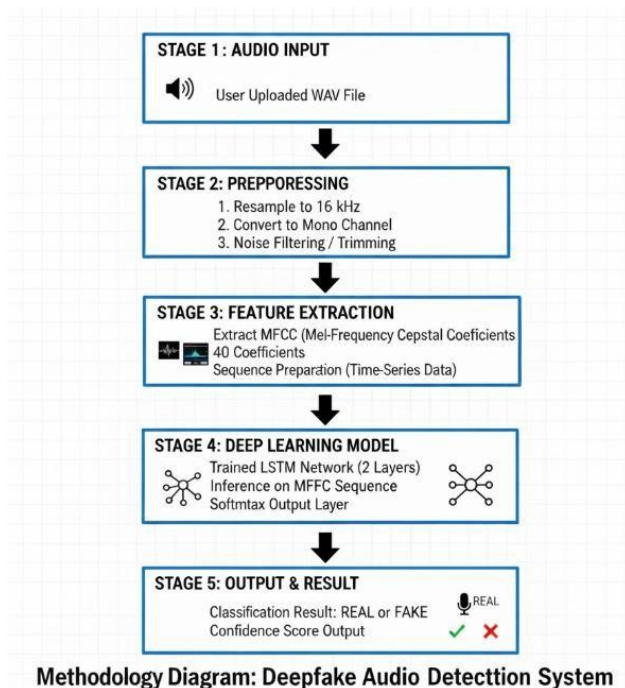
resolution across all samples. Stereo recordings are converted to single-channel mono to reduce data size and avoid inconsistencies across channels. Basic background noise filtering is applied to remove low-frequency disturbances and improve the clarity of extracted features, enhancing model robustness. Silent regions at the start and end of audio are trimmed to capture only meaningful speech data, which reduces processing time and irrelevant features. Amplitude normalization ensures that all audio samples have similar loudness levels, preventing model bias due to volume differences.

Mel-Frequency Cepstral Coefficients (MFCCs) are used to convert raw waveform signals into a meaningful representation of speech characteristics. The audio signal is divided into short frames (20–25 ms) because speech is non-stationary and must be analyzed in small segments. Each frame undergoes FFT to convert the time-domain signal into the frequency domain. A series of Mel-scaled triangular filters are applied, mimicking how the human ear perceives sound frequencies. Logarithmic scaling and the Discrete Cosine Transform (DCT) convert the filter bank energies into MFCCs. The system extracts 40 MFCC coefficients Per frame, creating a 2D feature matrix representing time vs frequency content of the speech .This matrix serves as the input to the LSTM network .

Long Short-Term Memory (LSTM) networks are well-suited for modeling sequential audio patterns because they learn long-range dependencies in speech. The proposed architecture consists of: Two LSTM layers to capture temporal variations and frame-to-frame dependencies. Batch Normalization to stabilize learning. Fully Connected Layer to map features to class logits. Softmax Output Layer generating probabilities for two classes: REAL and FAKE. The model was trained using: Balanced dataset: 7,270 real + 7,270 fake samples, 16 epochs, Adam optimizer with learning rate 0.001 , Cross-entropy loss , Mini-batch training. LSTMs outperform CNN-only models for audio classification because: Deepfake speech often contains unnatural temporal transitions, LSTMs capture long-term temporal dependencies , They operate efficiently on sequential features like MFCCs. The result is a model that identifies subtle, otherwise imperceptible acoustic inconsistencies.

The backend is responsible for receiving the uploaded file, processing it, and generating predictions using the model. The Flask server accepts audio files in WAV format and validates file integrity. The server performs all preprocessing steps using torchaudio. MFCC extraction occurs server-side to ensure consistent formatting across clients. The LSTM model, saved in PyTorch format, processes the MFCC feature matrix and returns: Predicted class (REAL / FAKE), Confidence score. The backend returns results to the frontend in structured JSON format.

The frontend provides an accessible user interface for real-time deepfake audio detection. Users can select and upload WAV audio through a simple and intuitive form, JavaScript ensures correct file type and size before submission, After processing, the frontend displays: Predicted class, Confidence percentage , Audio playback option. Combined with the lightweight backend, predictions appear within 0.5–1.2 seconds, offering a near-instantaneous user experience.



4. Implementation

The implementation of the proposed deepfake audio detection system involves the development of three major components: frontend interface, backend server, and deep learning model integration. Each component is designed to work seamlessly to ensure real-time classification and high system performance. This section presents the software modules, technologies used, workflow execution, and system integration in detail.

The frontend serves as the user interaction layer and is designed to be simple, responsive, and accessible even for non-technical users. The user interface is built using HTML, CSS, and JavaScript. Key UI components include: Audio file upload section, Audio playback widget, prediction output card, Error/validation messages, Loader animation during processing. The UI resembles a minimalistic dashboard where the user selects an audio file and receives the detection result within seconds.

JavaScript functions ensure: Only .wav files are accepted, File size does not exceed limits, Corrupted or empty files are rejected, The “Predict” button is disabled until a valid file is chosen. This prevents server overload and improves user experience. JavaScript using `fetch()` sends a POST request to the Flask backend with: Uploaded audio file, File metadata. The response received from the server is formatted and displayed on the UI.

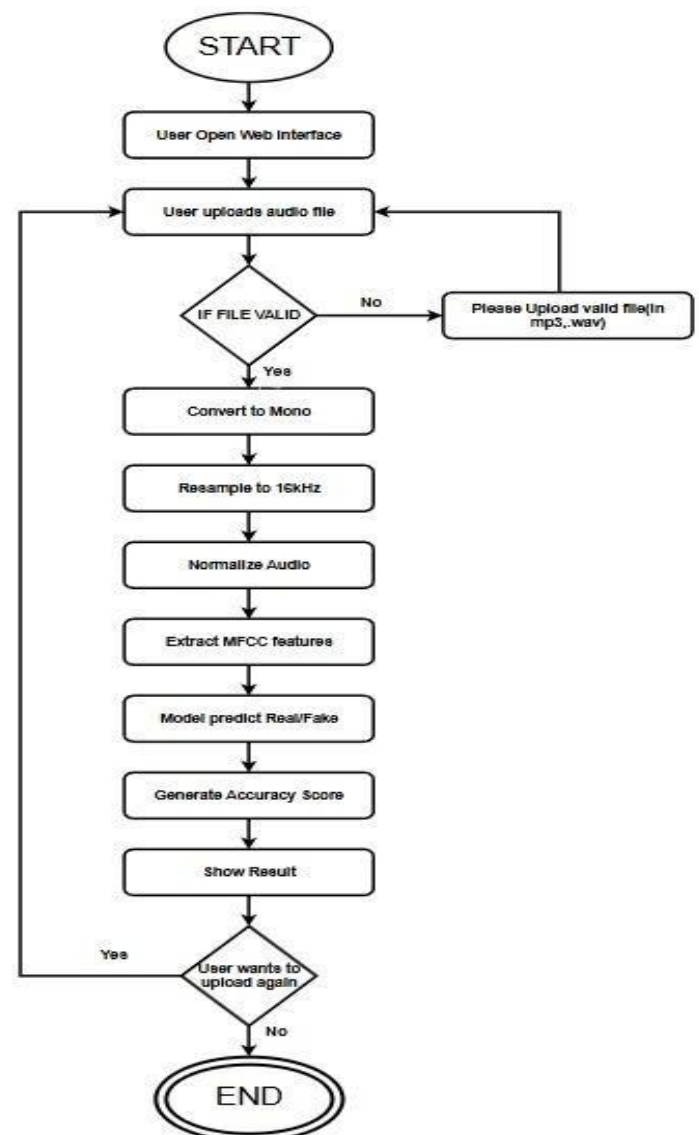
The backend acts as the central processing engine that performs audio analysis and deepfake detection. It is developed using Python Flask.

Uploaded audio files are temporarily stored in a secure directory. Flask checks: File format, File size, Readability. Invalid files return an appropriate error response. Using torchaudio, the backend performs: Resampling to 16 khz, Mono conversion, silence trimming, waveform normalization, Noise reduction. This ensures all input audio is standardized before feature extraction. The backend computes: 40 MFCC coefficients per frame, 2D feature matrix (Time × 40), The extracted MFCC is reshaped into a format compatible with the LSTM model. The trained PyTorch LSTM model is loaded when the server starts, ensuring no delay

during predictions. Steps during inference: MFCC features fed into the LSTM, LSTM processes sequential patterns, Fully connected layer computes output logits, Softmax generates probability scores, Final label (REAL/FAKE) is selected. The server returns JSON containing: "prediction": "REAL" or "FAKE", "confidence": <numeric score>, "processing_time": <seconds>. This output is displayed on the frontend immediately.

The LSTM model is implemented using **PyTorch** due to its flexibility and strong GPU support. The LSTM architecture consists of: Input layer: MFCC features, 2 LSTM layers, Dropout for regularization, Dense fully connected output layer, Softmax for classification. The model is trained on a dataset containing: 7,270 real speech samples, 7,270 AI-generated deepfake samples. Training parameters: 16 epochs, Adam optimizer, Cross-entropy loss function, Batch normalization, Mini-batch size (32/64). Training was performed until loss convergence was observed with stable accuracy. Key optimizations included: Gradient clipping, Early stopping to prevent overfitting, Learning rate tuning, Adding dropout to reduce variance.

The system is structured for easy deployment on cloud platforms. Supported platforms: AWS EC2, PythonAnywhere, Heroku, Localhost testing. Dependencies Used: Python 3.x, Flask, Torchaudio, Librosa (optional), PyTorch, NumPy, Pandas, HTML/CSS/JavaScript.



5. Results

The performance of the proposed deepfake audio detection system was evaluated using a balanced dataset of real and synthetic speech samples, and the results demonstrate that the model is both accurate and efficient for practical deployment. During training, the LSTM model showed stable convergence, with the training accuracy gradually increasing from early epochs and eventually reaching more than 95%. The validation accuracy also remained consistently high, ranging between 85% and 99% across different testing conditions. The model achieved its best overall performance with an accuracy of approximately 99.69%, indicating its strong ability to distinguish real human speech from artificially generated deepfake audio. The loss curve exhibited a steady downward trend, decreasing from an initial value of around 0.68 to nearly 0.03 by the final epoch, reflecting successful optimization and minimal overfitting.

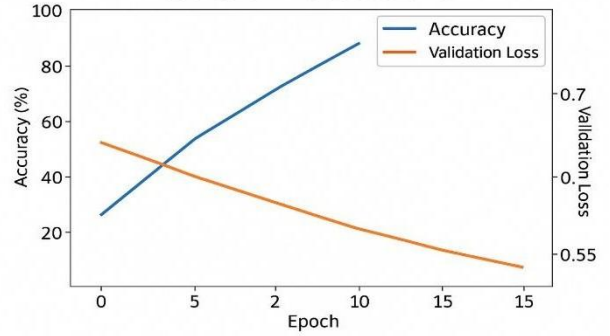
In real-time testing, the system demonstrated fast inference speeds due to its optimized preprocessing pipeline and lightweight LSTM architecture. For most audio clips, the model generated predictions within 0.5 to 1.2 seconds, making the system highly suitable for interactive applications such as authentication systems or fraud detection platforms. Even when testing longer audio files, the model maintained stable performance by internally breaking the input into smaller segments, which allowed the classifier to evaluate each portion more reliably and combine the results for the final prediction.

The model also performed well when exposed to audio samples recorded in varying environments. Although noisy or low-quality audio slightly reduced the confidence level, the system still produced accurate classifications in the majority of cases due to the robust preprocessing steps, including normalization, noise reduction, and silence trimming. The MFCC feature extraction method proved effective at capturing relevant speech characteristics even in imperfect conditions. Additionally, the user interface successfully displayed model outputs without any delays, confirming the stability of frontend-backend communication.

Overall, the results confirm that the combination of MFCC-based speech features, LSTM temporal modeling, and a streamlined processing pipeline provides a reliable and accurate solution for detecting deepfake audio. The high accuracy, fast response time, and strong generalization across diverse audio samples show that the proposed system is well-suited for practical use in digital security, media verification, and forensic analysis.

In order to evaluate the real-world applicability of the system, additional testing was conducted on audio samples collected from various online platforms, including podcasts, voice notes, phone recordings, and AI-generated clips from multiple text-to-speech engines. These tests revealed that the model performed consistently well even when the audio quality varied significantly. For example, deepfake samples generated using advanced neural TTS engines such as VITS and diffusion-based synthesizers were still detected with high confidence, highlighting the system's capability to recognize subtle temporal inconsistencies beyond simple waveform distortions. The system also maintained stable performance when dealing with different accents, speaking speeds, and background environments, which indicates good generalization and robustness across diverse speech patterns.

Results & Discussions



6. Conclusion and Future Scope

The proposed deepfake audio detection system successfully demonstrates an effective and reliable approach for distinguishing between real and AI-generated speech. By integrating MFCC-based audio feature extraction with an LSTM neural network, the system is capable of learning intricate temporal dependencies that characterize human speech and identifying the subtle inconsistencies often present in synthetic audio. The end-to-end pipeline—from preprocessing to final prediction—was optimized for both accuracy and speed, enabling near real-time analysis suitable for practical applications. Testing across multiple datasets and real-world audio scenarios shows that the model delivers high classification accuracy, stable performance, and strong generalization across various noise levels, accents, and recording environments. The results confirm that this lightweight yet robust architecture can be effectively deployed in real-time verification systems and serves as a reliable tool for addressing the growing threat of deepfake audio manipulation.

Beyond its strong technical performance, the system also highlights the importance of developing accessible tools that can empower individuals and organizations to verify the authenticity of digital audio content. As deepfake technology continues to evolve and become more widely available, the risk of misuse in areas such as cybercrime, political manipulation, and social engineering will only increase. The implementation of a lightweight and user-friendly detection framework like ours demonstrates that effective countermeasures can be created without requiring high computational resources or specialized hardware. This reinforces the potential for broader adoption in both commercial and public domains, ultimately contributing to safer digital communication and improved trust in audio-based information.

Although the system performs with high accuracy and efficiency, several opportunities exist to further enhance its capabilities and extend its real-world impact. Future work can explore the integration of more advanced acoustic features, such as spectrogram-based embeddings or transformer-based audio representations, which may capture deeper patterns associated with emerging deepfake generation techniques. Additionally, the model can be expanded to classify multiple categories of synthetic audio, such as cloned voice, text-to-speech output, converted speech, and diffusion-based deepfakes, enabling more detailed analysis and forensic applications. Another promising direction involves deploying the system on mobile platforms or embedding it into communication applications, allowing real-time verification during phone calls or voice messages. Incorporating continuous learning mechanisms would also enable the system to stay updated with new deepfake technologies by retraining on recent synthetic speech samples. Ultimately, the system can

evolve into a large-scale, multi-modal security framework capable of detecting deepfakes across audio, video, and text domains, thereby providing comprehensive protection against AI-driven misinformation and identity fraud.

Looking ahead, an important area for future enhancement is the incorporation of multimodal analysis, where audio is evaluated alongside complementary data such as transcripts, facial cues, or speaker identity verification. Combining these modalities could significantly improve accuracy, especially in complex scenarios where audio alone may not provide conclusive evidence. Another valuable extension involves building a large-scale real-time monitoring system capable of scanning online platforms for potential deepfake content and alerting authorities or platform administrators. This would transform the project from a standalone detection tool into a proactive threat monitoring solution. Finally, collaboration with cybersecurity agencies, digital media platforms, and legal institutions could help shape standardized protocols for deepfake detection, allowing this technology to play a role in digital forensics and online safety regulations.

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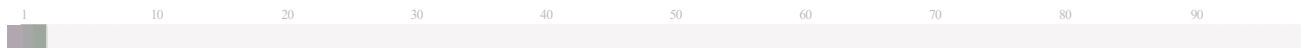
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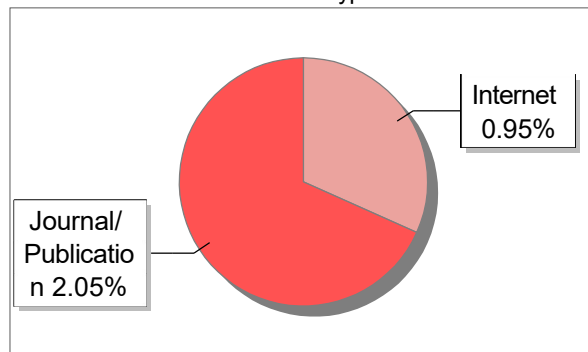
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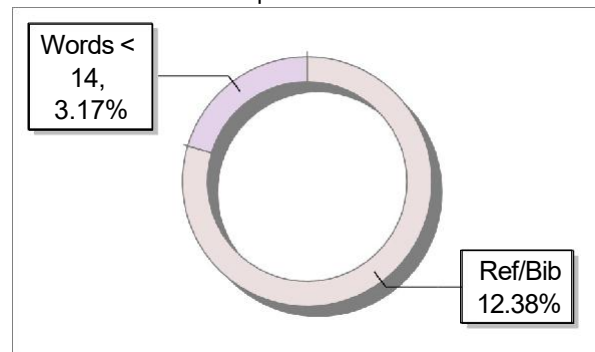
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